

Komal Jha

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Final-year Computer Science student at VIT Vellore who builds software across the stack, from real-time web apps and API infrastructure to machine-learning systems. Frontend internship at Tech Mahindra on an enterprise banking application. Projects are tested, load-checked and documented, with honest results.

Education

Vellore Institute of Technology — <i>B.Tech in Computer Science (CGPA: 7.51)</i>	2023 – 2027
Saint Francis School B.Deoghar, Deoghar (<i>Class X: 95.4%</i>)	2021
Swarajya Senior Secondary School, Alwar (<i>Class XII: 77.8%</i>)	2023

Experience

Tech Mahindra	May 2025 – July 2025
<i>Frontend Development Intern</i>	
- Contributed to frontend development of a Corporate Banking Solutions app for a PSU bank using Angular, HTML and CSS. - Improved UI consistency and responsiveness by resolving styling issues across existing application screens. - Collaborated with mentors and team members on enhancements aligned with business requirements, following Agile practices.	

Projects

GateKeeper – Multi-Tenant API Gateway & Rate Limiter Code · Live demo	2026
- Engineered a token-bucket rate limiter as an atomic Redis Lua script; load-tested so 20 concurrent requests always give the exact allowed / rejected split (no race conditions). - Built per-tenant monthly quotas in the same atomic Redis round-trip, plus SHA-256 hashed API keys (shown once) with instant revocation. - Built a live analytics dashboard (React, Recharts) for traffic and per-tenant quota usage. Tech: Node.js, Redis, PostgreSQL, React.	
SyncBoard – Real-Time Collaborative Kanban Board Code · Live demo	2026
- Built a MERN Kanban board with Socket.IO: drag-and-drop moves and live cursors update instantly for all viewers. - Implemented 3-tier role-based access control enforced on both REST and WebSocket layers, closing a live-session access gap. - Designed the UI from scratch with a custom design system; deployed on Render and Vercel.	
StreamVault – Video Upload & Streaming Platform Code	2026
- Built an upload pipeline with background FFmpeg transcoding into 360p / 480p / 720p renditions, stored in MongoDB GridFS, with range-request streaming and adaptive quality. - Designed a watch-time analytics dashboard showing viewer drop-off; 16 end-to-end tests pass. Tech: React, Node.js, MongoDB.	
Dental X-ray Caries Classifier (ML) Code · Live demo	2026
- Fine-tuned an ImageNet ResNet-18 on 5,000 panoramic X-rays: ROC-AUC 0.74 (95% CI 0.70 to 0.77) and 65.9% accuracy on a held-out set of 750 unseen images, with bootstrap confidence intervals. - Implemented Grad-CAM from scratch for explainability; audited the data (no healthy images, one source shortcut removed) and documented limitations. Tech: PyTorch, Gradio.	
Heart Disease Prediction Pipeline (ML) Code · Live demo	2026
- Built a leak-free scikit-learn pipeline and compared 5 models under identical repeated stratified CV; best (logistic regression) ROC-AUC 0.92 ± 0.04, accuracy 85.8%. Tuned the top 2 with nested CV and explained results with permutation importance. - Reported negative results honestly (tuning and engineered features gave no gain). Tech: Python, scikit-learn, XGBoost, Pandas.	
CareerVault – Job Application & Career Management Platform Code	2026
- Built a MERN platform tracking job applications and interviews, with a Recharts dashboard, search, filters and pagination. - Implemented JWT authentication, bcrypt password hashing and RESTful CRUD APIs on MongoDB Atlas.	

Skills

- Languages:** Java, Python, JavaScript, SQL, C++, C
- Frameworks & Libraries:** React.js, Node.js, Express.js, Angular, FastAPI, PyTorch, scikit-learn
- Databases & Storage:** MongoDB, PostgreSQL, Redis
- Tools & Platforms:** Git, Postman, Socket.IO, Render, Vercel, MongoDB Atlas
- Core Concepts:** Data Structures & Algorithms, REST APIs, JWT Authentication, WebSockets, System Design, Concurrency, Rate Limiting, RBAC, Transfer Learning

Certifications

AWS Certified Cloud Practitioner — Amazon Web Services	2026
Data Structures & Algorithms — Udemy	2026